OJT – Project Proposal (PRD)

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Track: Product Development
Project Title: Loci - An Audio-First Spatial Guide for Structured B2B Environments

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1. Problem Understanding

1.1 What is the problem statement in your own words?

People moving through large structured environments such as **campuses, museums, and institutional buildings** often pass through meaningful locations without understanding their context or significance. While information exists, accessing it usually requires stopping, reading signage, or interacting with screens, which interrupts natural movement and presence.

Existing digital solutions are either screen-heavy or unreliable indoors, leading to poor timing, incorrect information, or excessive user interaction. There is no lightweight, audio-first system that provides spatial context while people are in motion, without demanding constant attention.

1.2 Why does this problem exist or matter? (WHY Component)

This problem exists because most navigation and tour systems are designed with a **precision-first approach**, assuming that accurate real-time location data is always available. In indoor and semi-indoor environments, this assumption fails due to unreliable GPS and inconsistent sensor signals.

As a result, systems either guess the user's location - leading to incorrect guidance - or rely heavily on visual interfaces, increasing cognitive load. In **B2B environments**, such as campuses, museums, and office buildings, incorrect or misleading information reduces trust and credibility.

This problem matters because these environments are **structured, long-lived, and visited repeatedly by new users**. A system that prioritizes correctness over constant feedback, and treats **silence as better than wrong information**, can deliver a calmer and more reliable experience while remaining usable across different types of structured physical spaces.

1.3 Key inputs and expected outputs

Inputs	Process	Expected Outputs
➢ User movement within a structured environment	➢ Confidence-based spatial reasoning	➢ Contextual audio narration
➢ Predefined spatial data (buildings, floors, zones)	➢ Offline routing & validation	➢ Correctly timed guidance
➢ Curated audio content	➢ Confidence-gated playback	➢ Silence when unsure
➢ Optional voice queries	➢ Controlled AI retrieval	➢ Grounded responses

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2. Functional Scope

2.1 What are the core features you plan to build (must-haves)?

- Audio-first spatial guidance
- Support for structured B2B environments (campuses, museums, buildings)
- Confidence-based audio triggering (no forced autoplay)
- Indoor and outdoor context handling
- Offline audio and map caching
- Deterministic routing using predefined paths
- Controlled AI-based Q&A (optional, bounded)
- Minimal UI for status and prompts
- Prevention of incorrect or overlapping audio

2.2 What stretch goals could you attempt if time permits?

- Exploratory routing (landmark-based paths)
- Improved indoor-outdoor transition handling
- Multi-floor expansion for larger environments
- Limited analytics for evaluation and testing

2.3 Which libraries or tools will you use?

- React Native
- Node.js with TypeScript
- Offline storage (cached location packs)
- Cloud LLM with retrieval-based access

● Web-based internal builder tool
● Git & GitHub

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3. System & Design Thinking

3.1 Sketch or describe your app flow / pipeline

User opens the app → selects a location
→ App loads structured location data (offline)
→ User moves naturally through the environment
→ System evaluates confidence of proximity
→ Audio plays only when confidence is high
→ System remains silent when confidence is low

Flow Summary:
Structured data → confidence reasoning → intentional audio → silence when unsure

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3.2 What data structures or algorithms are central to this project?

● Spatial hierarchy: Location → Building → Floor → Zone / POI
● Walkable path graph (nodes and edges)
● Confidence states: low / medium / high
● Audio metadata linked to zones
● Knowledge capsules for AI responses

3.3 How will you test correctness or performance?

● Walkthrough testing across predefined routes
● Verifying audio only triggers at correct confidence
● Testing silence behavior when confidence drops
● Offline testing without network access
● Repeated runs for consistency

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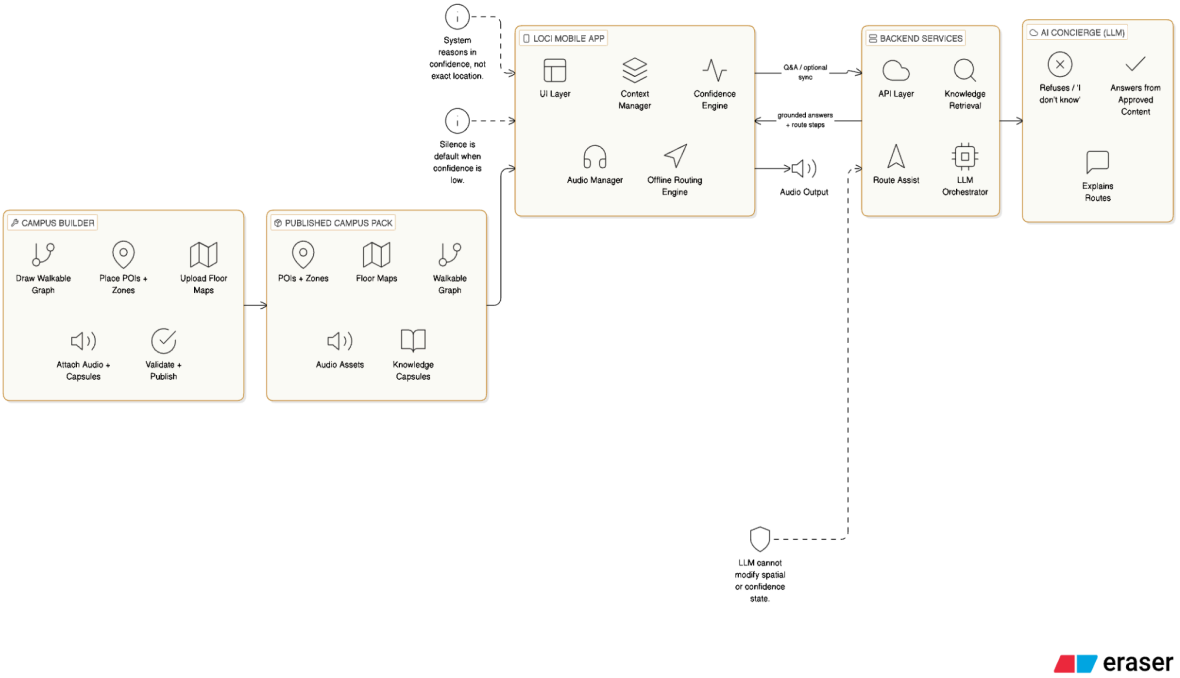
4. High Level Design (HLD)

The system is designed as a **data-driven architecture** where structured environments are authored once and consumed by the mobile app at runtime.

Major Components:

- **Internal Builder Tool:** Used to define maps, paths, zones, audio, and approved knowledge.
- **Published Location Pack:** Versioned, offline-first data bundle consumed by the app.
- **Mobile Application:** Handles UI, confidence reasoning, routing, and audio playback.
- **Backend Services:** Provides APIs and controlled AI orchestration.
- **AI Layer:** Answers questions strictly from approved content and cannot modify spatial or confidence state.



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5. Timeline & Milestones (12 Weeks)

Phase	Duration	Milestones / Deliverables
Phase 1: Planning & Design	Week 1–2	Finalize requirements, lock PRD and HLD, define spatial model and confidence logic
Phase 2: Builder Tool v1	Week 3–4	Develop internal builder tool with map upload, path drawing, POI placement, and data export

Phase 3: Mobile App Core	Week 5–6	Set up React Native app, load published location packs, basic UI and navigation
Phase 4: Confidence & Audio Logic	Week 7–8	Implement confidence-based reasoning, audio triggering, silence handling, offline playback
Phase 5: Routing & AI Integration	Week 9–10	Implement graph-based routing and controlled AI Q&A with strict boundaries
Phase 6: Testing & Demo Preparation	Week 11–12	End-to-end testing, offline demo setup, documentation, final presentation

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6. Risks & Dependencies

6.1 What's the hardest part technically for you right now?

- Designing confidence-based behavior without exact indoor positioning
- Preventing incorrect audio playback
- Enforcing strict AI boundaries

6.2 What dependencies or help do you need from mentors?

- Feedback on system boundaries
- Validation of architectural decisions
- Review of evaluation readiness

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7. Evaluation Readiness

7.1 How will you prove that your project "works"?

- Live or recorded demo in a structured environment
- Screenshots of the internal builder and mobile app

- Offline demo walkthrough
- Code repository with documentation

7.2 What success metric or goal will you aim for?

- No incorrect audio playback
- Clear indoor–outdoor transitions
- Controlled and grounded AI responses
- Minimal user interaction required

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